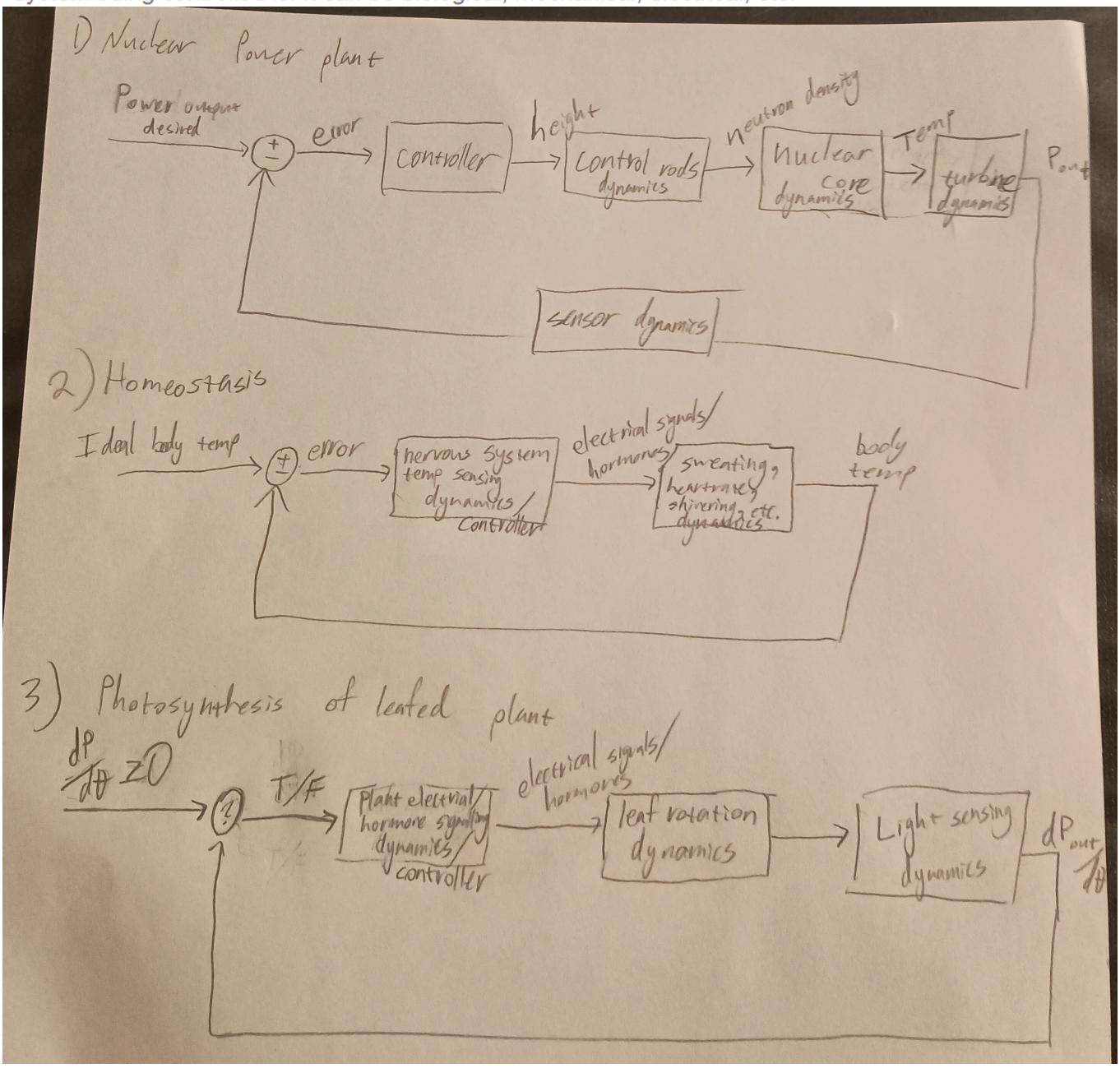
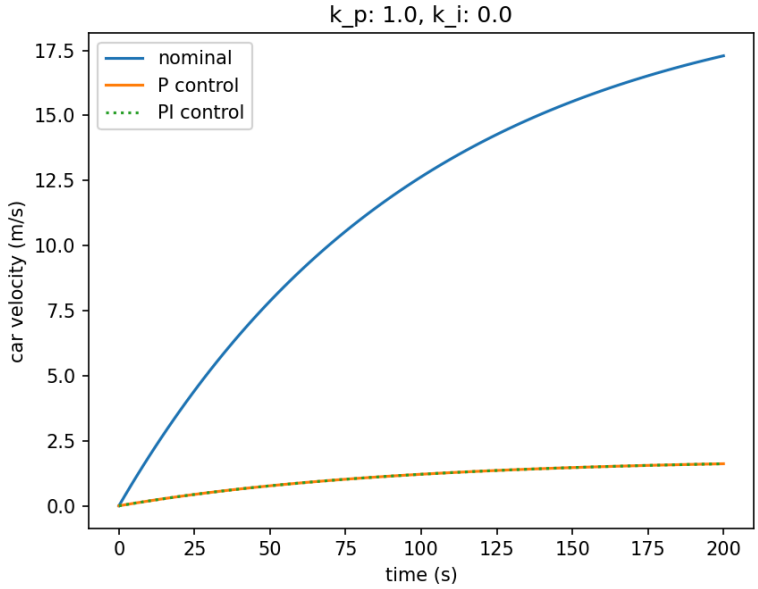
Assignment 1

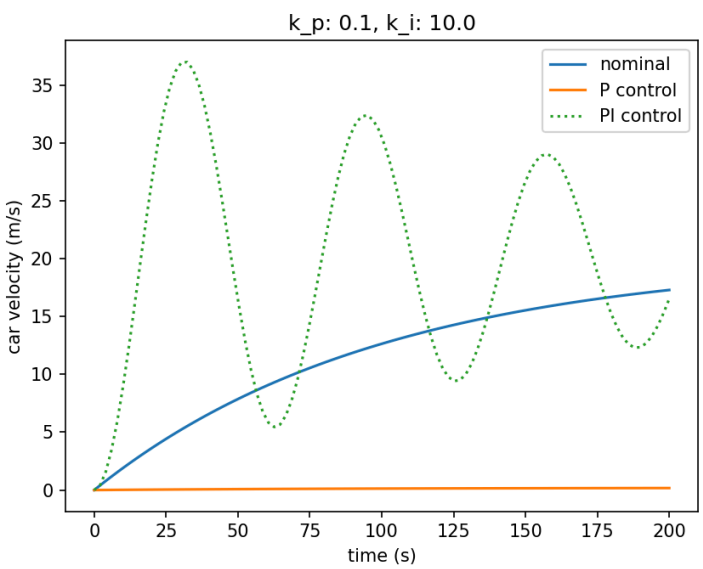
(4 pts) List 3 different control systems you see in the world around you. Draw a picture that represents each control system as a block diagram, and make sure to clearly say what the actuators, sensors, and system being controlled is. It can be biological, mechanical, electrical, etc.



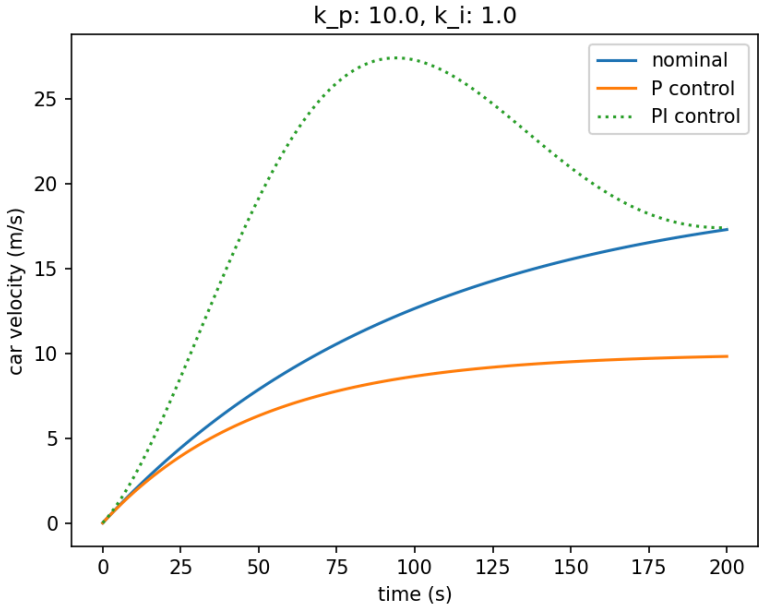
(8 pts) Spend at least 10-15 minutes playing with the parameters k_p, k_i, and u_cmd in the code below for case study 0 (cruise control example). For every pair of gains you pick, please make sure you know what the resulting poles are. Try to build up some intuition about poles and how they affect the time response based on the behavior you see in the plots and how that relates to the linearized time solution for the cruise control system. Turn in at least two different plots (or more) that show what you learned about the trends as you change the two available gains.



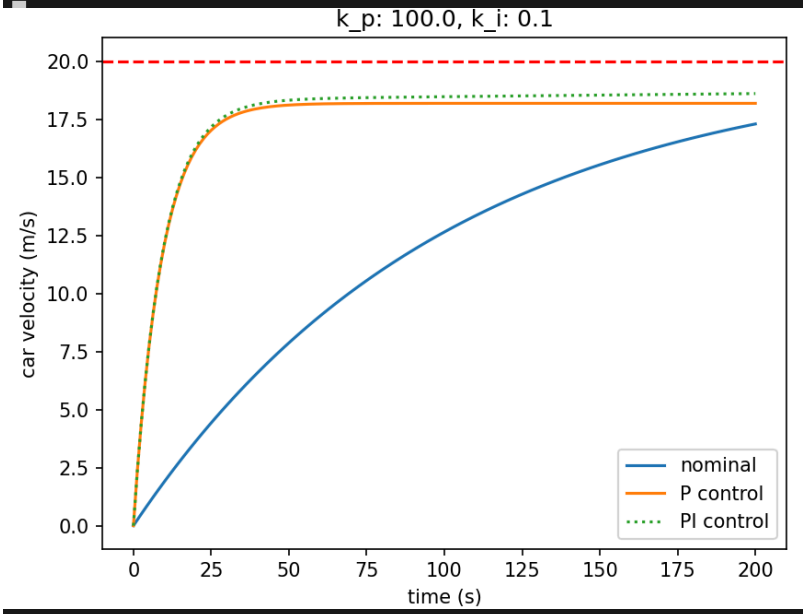

```
roots of each system are:
for open-loop: [-0.01+0.j]
for closed-loop with kp: [-0.011+0.j]
for closed-loop with kp and ki: [-0.011+0.j  0.   +0.j]
```



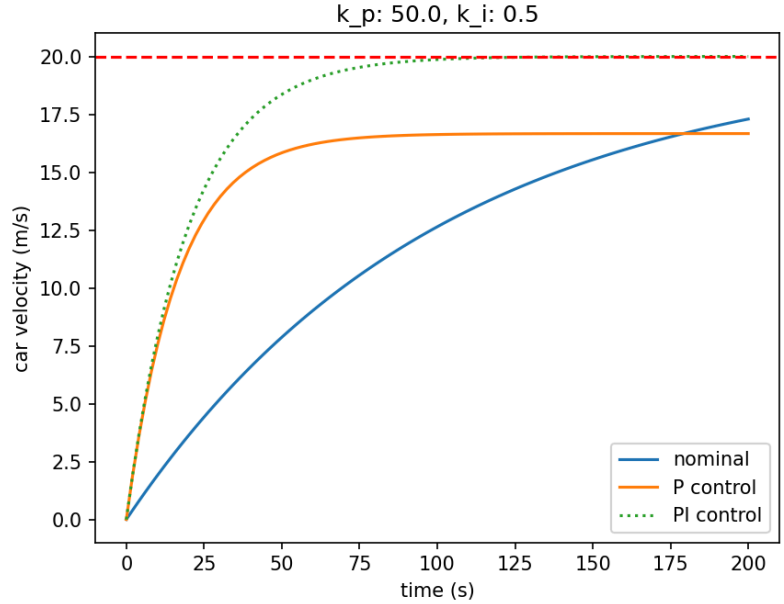

```
roots of each system are:
for open-loop: [-0.01+0.j]
for closed-loop with kp: [-0.0101+0.j]
for closed-loop with kp and ki: [-0.00505+0.09987241j -0.00505-0.09987241j]
```




```
roots of each system are:
for open-loop: [-0.01+0.j]
for closed-loop with kp: [-0.02+0.j]
for closed-loop with kp and ki: [-0.01+0.03j -0.01-0.03j]
```

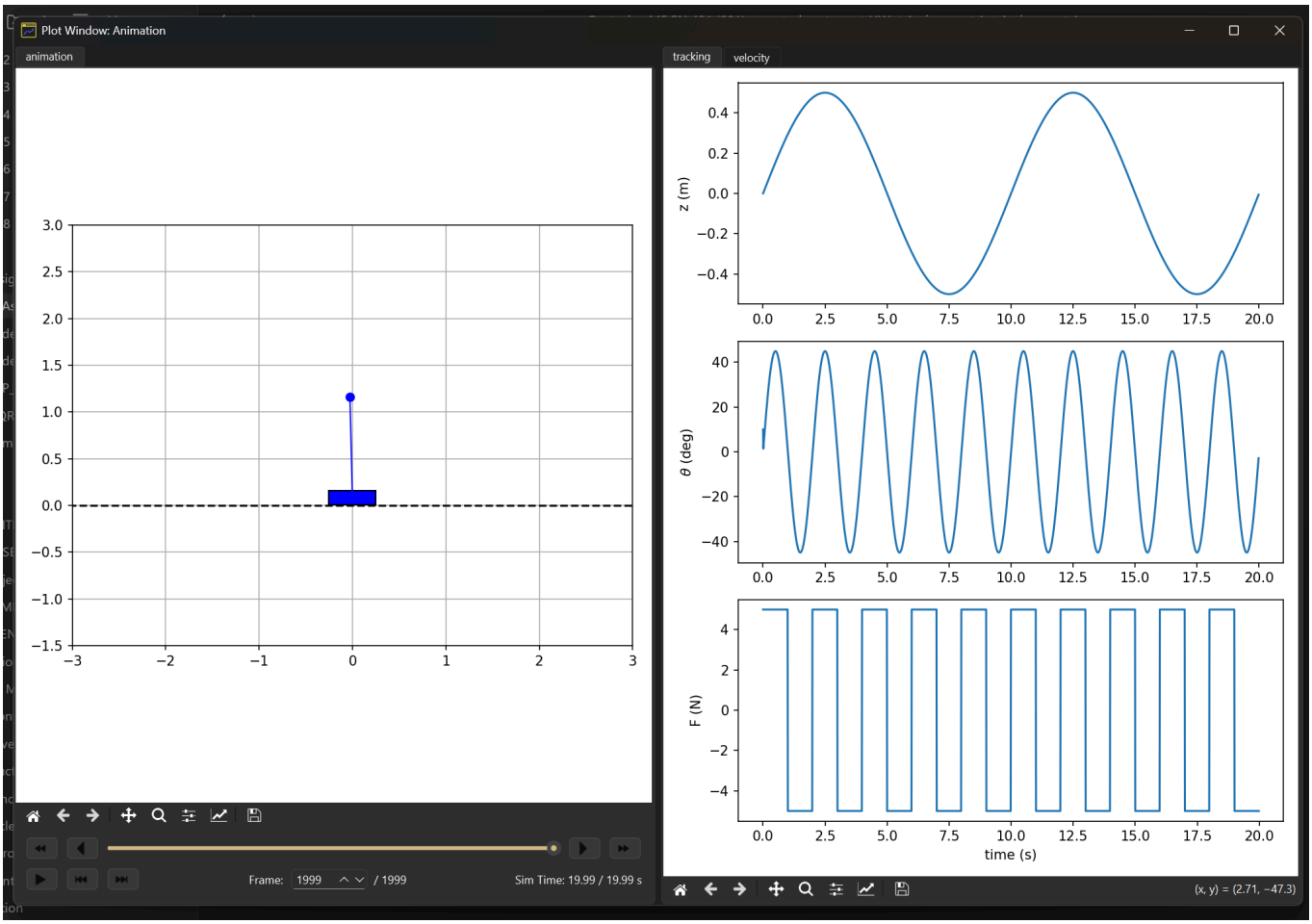



```
roots of each system are:
for open-loop: [-0.01+0.j]
for closed-loop with kp: [-0.11+0.j]
for closed-loop with kp and ki: [-0.10908327+0.j -0.00091673+0.j]
```




```
roots of each system are:
for open-loop: [-0.01+0.j]
for closed-loop with kp: [-0.06+0.j]
for closed-loop with kp and ki: [-0.05+0.j -0.01+0.j]
```

(8 pts) Complete the Python tutorial code on Learning Suite if you are unfamiliar with Python. Run each cell and making sure you understand the output. Then update and fix debug01.py and debug02.py to help you understand how to debug and interpret Python code. Each file is worth 4 points.

Done.

(4 pts) Download and install the class repository. Run at least one file from the public case studies (in chap02, or chap03, or chap08, etc.). If it installs and runs correctly, you can give yourself 4 points.